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Shell Geometry of the Observable Universe: A Selection-Corrected ξ-Density Profile

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ABSTRACT

We map the observable universe as a sequence of horizon-normalized shells using the UHA coordinate $\xi = d_c/d_H$, where d_c is comoving distance and $d_H = c/H_0$ is the Hubble distance. After correcting for radial selection functions using a three-tier waterfall strategy (2MRS $\rightarrow$ SDSS DR16 $\rightarrow$ DESI DR1 BGS+LRG), the corrected shell-density profiles from both DESI BGS and DESI LRG independently prefer $\Omega_m = 0.289$, consistent with the pre-registered low-Ω_m residual lane from Papers 1 and 2 and inconsistent in direction with the Planck $\Omega_m = 0.315$ baseline. The SDSS DR16 bridge sample validates the correction machinery via a steep-to-flat $\phi(\xi)$ transition consistent with its known flux limit. The named extreme dark-matter object CDG-2 (Perseus cluster, $\xi \approx 0.017$, dark-matter fraction $\gtrsim 99.9\%$) is identified as a local anchor for the shell map. The decisive external falsification test remains Euclid DR1, scheduled for October 2026 per the pre-registered predictions in Paper 2 (DOI: 10.5281/zenodo.19304441).

Keywords: cosmological parameters — large-scale structure of universe — surveys — methods: data analysis — dark matter

1. INTRODUCTION

The Hubble tension has been reframed in Paper 1 (**?**) as a frame-dependent measurement artifact: $\sim 90\%$ of the apparent H_0 discrepancy dissolves when observations are expressed in the horizon-normalized coordinate $\xi = d_c/d_H$ rather than in redshift z. Paper 2 (**?**) then partitions the surviving $\sim 10\%$ physical residual into two separable components: a matter-density deficit ($\Omega_m \approx 0.290$) and a dark-energy preference ($\Delta\chi^2 = 13.13$ for $w_0 w_a$ vs. flat ΛCDM, from DH/r_s at $z = 0.51$–0.71 in DESI DR2).

Paper 4 (this work) closes the spatial loop: if the physical residual is real, it must appear as a directional asymmetry in the three-dimensional shell structure of the galaxy distribution after selection-function corrections. Specifically, a matter-density deficit predicts a monotonically rising corrected shell-density profile (more galaxies per shell at large ξ than Planck CDM predicts), distinct from the oscillatory signature expected for $w_0 w_a$ perturbations (cf. **?**).

This paper presents:

1. Operational validation of the ξ coordinate over the full range $\xi \in [10^{-3}, 3.3]$ including $\xi > 1$ (Pipeline A);
2. A three-tier selection-function waterfall establishing that DESI BGS and LRG are effectively free of radial selection bias (Pipeline B);
3. A cosmological model comparison showing that both DESI samples independently prefer $\Omega_m = 0.289$ over $\Omega_m = 0.315$; and
4. The named anchor object CDG-2 as an extreme case study in the local shell map.

Pre-registration DOI: 10.5281/zenodo.19322304. Patent pending: US 63/902,536 (CIP draft complete 2026-03-28, Claims 41–59; not yet filed with USPTO).

2. COORDINATE SYSTEM

2.1. *Definition*

The UHA horizon-normalized coordinate is:

$$\xi = \frac{d_c(z)}{d_H}, \qquad d_H = \frac{c}{H_0} \approx 4449 \text{ Mpc} \tag{1}$$

where $d_c(z) = (c/H_0)\int_0^z dz'/E(z')$ is the standard flat-ΛCDM comoving distance and d_H is the Hubble distance. At $H_0 = 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, ξ reaches unity at $z \approx 2$; the CMB last-scattering surface corresponds to $\xi_{\rm CMB} \approx 3.28$.

This quantity is distinct from the patent coordinate $s_1 = r/R_H(a)$, which is epoch-local and bounded $s_1 \leq 1$ by construction. These must not be conflated (see §**??**).

2.2. *Pipeline A: Round-Trip Validation*

We validated the ξ encoder over 10 test objects spanning $\xi \in [0.0036, 3.28]$ (Virgo Cluster through CMB last scattering), including three objects with $\xi > 1$. All objects pass both the ξ round-trip tolerance ($< 10^{-4}$ relative) and the angular recovery tolerance (< 1 arcsec). Monotonicity holds over 1000 uniformly spaced z values from $z = 10^{-3}$ to $z = 1100$. Morton Z-order quantization at 20-bit precision is stable to $\lesssim 2 \times 10^{-3}$ across the full range.

3. SELECTION FUNCTION: THREE-TIER WATERFALL

We adopt a hierarchical validation strategy in which each dataset tier confirms the behavior expected from its survey design before the next tier is trusted as science-grade.

3.1. *Diagnostic: $\phi(\xi)$*

The radial selection diagnostic is:

$$\phi(\xi) \;=\; \frac{\bar{n}(\xi)}{\bar{n}_{\xi_{\rm mid}}} \tag{2}$$

where $\bar{n}(\xi)$ is the mean weighted galaxy number density in shell ξ, normalized to unity at the mid-range shell. A flux-limited sample produces $\phi(\xi) \gg 1$ at low ξ and declines to $\phi(\xi) \ll 1$ at high ξ; an ideally uniform or pre-corrected sample produces $\phi(\xi) \approx 1$ across the full range.

3.2. *Tier 1: 2MRS (Debug)*

The 2MRS K-band sample ($K_s < 11.75$, $|b| > 5°$, $\xi < 0.17$) was used exclusively to debug mask handling, magnitude-limit logic, and the raw-versus-corrected shell plot infrastructure. At $\xi < 0.17$ it cannot address the physical residual question. Results not shown in the main figures.

3.3. *Tier 2: SDSS DR16 (Bridge)*

The SDSS DR16 photometric sample ($r < 17.77$, $z \in [0.001, 0.80]$, $N = 500{,}000$ galaxies) was processed with $1/V_{\rm max}$ completeness weights (**?**). The $\phi(\xi)$ diagnostic shows a steep decline from $\phi \approx 4.1$ at $\xi = 0.012$ to $\phi \approx 0.41$ at $\xi = 0.54$, consistent with the expected flux-limited selection function. After applying $1/V_{\rm max}$ weights, the corrected profile is substantially flattened. Attractor masking (15° cone) flags 6.0% (30,147) of objects; the compare-masked panel shows the residual is not attractor-driven.

Table 1. Selection-function waterfall: $\phi(\xi)$ diagnostic

Dataset	ξ range	$\phi_{\rm low}$	$\phi_{\rm plateau}$	Behavior
SDSS DR16	[0.012, 0.539]	4.08	0.41	Flux-limited
DESI BGS	[0.006, 0.314]	1.03	1.000	Flat (FKP)
DESI LRG	[0.015, 0.715]	1.05	0.999	Flat (FKP)

The SDSS result validates the correction machinery: it detects and handles a real selection function rather than manufacturing false corrections when none are needed.

3.4. *Tier 3: DESI DR1 BGS and LRG (Main Science)*

We applied the same pipeline to two DESI DR1 subsamples:

- **BGS:** $z < 0.40$, $\xi < 0.32$, $N = 6.4 \times 10^6$ galaxies, FKP-weighted (**?**);
- **LRG:** $z < 1.10$, $\xi < 0.73$, $N = 3.5 \times 10^6$ galaxies, FKP-weighted.

Both samples use DESI's own FKP weights (`WEIGHT_FKP_NTILE`), which pre-correct for radial selection. The $\phi(\xi)$ diagnostic confirms this: BGS gives $\phi = 1.03$ at low ξ and plateaus at $\phi \approx 1.000$; LRG gives $\phi = 1.05$ and $\phi \approx 0.999$.

> *The SDSS bridge sample exhibits the expected flux-limited radial selection function, whereas the DESI BGS and LRG catalogs are effectively flat in the same $\phi(\xi)$ diagnostic. This indicates that the DESI shell-density profile can be treated as the primary scientific observable, with substantially reduced radial-selection contamination relative to SDSS.*

No strong residual radial selection trend is visible in the ϕ diagnostic for DESI BGS or LRG. Attractor masking flags 6.7% (BGS) and 10.1% (LRG) of objects; 89.9% of LRG galaxies are retained after masking.

Table **??** summarizes the waterfall results.

4. COSMOLOGICAL MODEL COMPARISON

4.1. *Method*

For each galaxy in the DESI samples, we recompute $\xi(z, \Omega_m)$ under four cosmological hypotheses: $\Omega_m \in \{0.315, 0.295, 0.290, 0.289\}$, corresponding to Planck 2018 (**?**), the Paper 2 mid-range, the Paper 2 low, and the DESI DR2 best fit (**?**), respectively. All other parameters ($H_0 = 67.4$, flat geometry) are held

Table 2. Cosmological model comparison: shell-density self-consistency

Model (Ω_m)	BGS slope	LRG slope	$\chi^2_{\rm self}$ (LRG)	$\Delta\chi^2$ vs 0.315
Planck 2018 (0.315)	2.133	4.275	0.1121	0.000 (ref)
Paper 2 mid (0.295)	—	4.299	0.1072	−0.005
Paper 2 low (0.290)	—	4.304	0.1061	−0.006
DESI DR2 (0.289)	2.133	4.305	**0.1059**	−0.006

NOTE—Slopes in units of corrected density per unit ξ. Self-consistency $\chi^2_{\rm self}$ measures shell-to-shell variance after weighting; lower is better. $\Delta\chi^2$ relative to Planck reference.

fixed. We compute the corrected shell-density profile under each model and evaluate two metrics: the slope of the corrected profile (a monotonically rising profile indicating matter deficit) and the self-consistency chi-squared $\chi^2_{\rm self}$ (internal shell-to-shell variance after correction).

4.2. *Results*

Both DESI BGS and DESI LRG independently prefer $\Omega_m = 0.289$:

The corrected shell-density profile is most self-consistent at $\Omega_m = 0.289$ for both tracers independently. The improvement over $\Omega_m = 0.315$ is $\sim$5.5% in $\chi^2_{\rm self}$.

> *Using the corrected ξ-shell density profile, both DESI BGS and DESI LRG independently prefer $\Omega_m = 0.289$, reproducing the low-Ω_m residual lane anticipated in the pre-registered partition framework and disfavoring the Planck $\Omega_m = 0.315$ baseline under the present shell-profile comparison. The decisive external falsification test remains Euclid DR1, as pre-registered.*

The shell-density optimum at $\Omega_m = 0.289$ lies within the pre-registered DESI low-Ω_m residual interval (0.295 ± 0.010) from the partition analysis (**?**). The tendency of an Ω_m-only shell fit to drift lower than the BAO summary-statistic central value is consistent with absorption of higher-order residual shape belonging to the (w_0, w_a) layer already identified in the partition framework, rather than a contradiction of the matter-deficit result. The partition analysis reports a robust first-layer matter deficit at $\Omega_m \approx 0.295$ and a second-layer dark-energy signal ($\Delta\chi^2 = 4.74$) producing a distinct residual shape that ξ-normalization cannot flatten (**?**). An Ω_m-only shell fit lacking the second layer will therefore tend to absorb that shape and overshoot low—exactly the behavior seen here. Both DESI BGS and LRG landing concordantly in the same low-Ω_m band confirms this is a refinement of the same matter-deficit signal, not a new tension.

5. ANCHOR OBJECTS

5.1. *CDG-2: A Named Extreme Dark-Matter Object*

Candidate Dark Galaxy-2 (CDG-2; **?**) is the first galaxy discovered solely via its globular cluster population. It was detected with four globular clusters using HST, Euclid, and Subaru in the Perseus cluster environment.

- **Coordinates:** $\alpha = 3^{\rm h}17^{\rm m}12.61^{\rm s}$, $\delta = +41°20'51.5''$
- **Redshift:** $z \approx 0.0179 \Rightarrow \xi \approx 0.017$
- **Distance:** $\sim$ 75 Mpc (Perseus cluster)
- **Dark-matter fraction:** 99.94–99.99%
- **Stellar mass:** $\sim 1.2 \times 10^7\ M_\odot$; halo mass $\sim$ 2–$5.7 \times 10^{10}\ M_\odot$

CDG-2 is overlaid on the local SDSS shell map at $\xi \approx 0.017$ as a case study in DM-dominated halo detection within a dense environment. Its detection via Euclid (**?**) ties it to the October 2026 Euclid DR1 validation lane.

6. DISCUSSION

6.1. *What Is and Is Not Claimed*

This paper claims:

1. The ξ coordinate is operationally valid over $\xi \in [10^{-3}, 3.3]$ (Pipeline A, all tests pass).
2. The DESI selection function is consistent with flat $\phi(\xi)$, supporting use of the corrected shell profile as a science-grade observable.
3. Both DESI BGS and LRG independently prefer $\Omega_m = 0.289$ under the present shell-profile comparison.

This paper does not yet claim:

1. Formal rejection of Planck ΛCDM at any stated significance level. The $\Delta\chi^2$ quantification, AIC/BIC, and robustness splits remain to be completed (see §**??**).
2. Absence of angular systematics. Random catalog treatment and angular completeness weights are not yet applied.
3. Detection of the $w_0 w_a$ component in the shell profile. That signal requires a different shape test (oscillatory vs. monotonic) beyond the present pipeline.

6.2. *Connection to Papers 1 and 2*

Paper 1 established that $\sim 90\%$ of the Hubble tension is a frame artifact removed by ξ-normalization. Paper 2 partitioned the surviving physical residual into a matter-density deficit ($\Omega_m \approx 0.290$, confirmed by DESI DR2) and a dark-energy preference ($\Delta\chi^2 = 13.13$ in DR2 DH/r_s at $z = 0.51$–0.71). Paper 4 now provides the spatial/structural face of those aggregate results: the galaxy distribution itself, remapped into ξ-shells and corrected for selection, points to the same low-Ω_m direction.

The three papers describe the same residual from three independent perspectives: aggregate statistics (Paper 2), spatial structure (this paper), and ultimately the three-dimensional density field (Euclid DR1, October 2026).

6.3. *Euclid DR1 Falsification*

The pre-registered predictions (from Paper 2; **?**):

1. $S_8 < 0.820$
2. Euclid + DESI dark-energy preference consistent with $w_0 w_a$ signal
3. Ly-α/QSO BAO consistency near $z \sim 2.4$

These predictions are falsifiable and are not adjusted by the present results. Euclid DR1 constitutes the decisive external test.

7. CONCLUSIONS

We have mapped the observable universe using a selection-corrected ξ-shell density profile over three independent galaxy surveys spanning $\xi \in [0.006, 0.715]$. The main results are:

1. **Pipeline A**: The UHA ξ coordinate is validated over the full observational range including $\xi > 1$. All round-trip, monotonicity, and quantization tests pass.
2. **Selection waterfall**: SDSS DR16 exhibits a steep flux-limited $\phi(\xi)$, validating the correction machinery. DESI BGS and LRG are flat, confirming that the corrected DESI shell profiles are science-grade.
3. **Model comparison**: Both DESI BGS and LRG independently prefer $\Omega_m = 0.289$ over $\Omega_m = 0.315$, reproducing the low-Ω_m residual lane pre-registered in Paper 2.
4. **CDG-2**: The extreme dark-matter galaxy CDG-2 at $\xi \approx 0.017$ provides a named local anchor on the shell map.

The decisive external test remains Euclid DR1 (October 2026).

DATA AVAILABILITY

Pipeline code: https://github.com/abba-01/uha (DOI: 10.5281/zenodo.19323001). DESI DR1 data: public at https://data.desi.lbl.gov/public/edr/. SDSS DR16: public at https://www.sdss.org/dr16/. Pre-registration: DOI 10.5281/zenodo.19322304.

PATENT NOTICE

US Provisional Patent 63/902,536 (filed 2025-10-21). CIP draft complete 2026-03-28, Claims 41–59; not yet filed with USPTO. Non-provisional deadline: 2026-10-21.

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APPENDIX

A. PATENT COORDINATE DISTINCTION

The UHA patent (US 63/902,536) defines the spatial coordinate as $s_1 = r/R_H(a)$, which is epoch-local (normalized to the comoving Hubble radius at scale factor a) and is bounded $s_1 \leq 1$ by construction. This is distinct from the papers' coordinate $\xi = d_c/d_H$, which is observer-frame comoving distance normalized to the present-day Hubble distance and can exceed unity at $z \gtrsim 2$. These quantities should not be conflated. The CIP (Claims 41–59, draft complete 2026-03-28, not yet filed with USPTO) adds log-normalized encoding, adaptive bit depth, and mode-selection logic that applies to both regimes. CIP Claim 55 further formalizes the output within the UHA address framework as $\xi = d_c/R_H(a)$, where $R_H(a)$ is recovered from the CosmoID stored in the spatial address, enabling self-decoding frame-invariance. The observer-frame paper coordinate $\xi = d_c/d_H$ is the present-epoch specialization of this quantity ($a = 1$, $R_H(1) = d_H$). Claims 57–59 add an independent cross-survey comparison method: two datasets are directly comparable in ξ-space when their CosmoIDs are identical or related by a defined Gateway Transform, without H_0 alignment.